

SUBRESULTANTS IN ROOTS WITHOUT DENOMINATORS

ABSTRACT. The formulas of D’Andrea, Krick and Szanto express subresultants of univariate polynomials as a ratio of two determinants, the denominator being a Vandermonde determinant in the roots. We show that the numerator is always divisible by the Vandermonde determinant over $\mathbb{Z}$, and we describe the quotient explicitly. The quotient is a multi-Schur function in the sense of Lascoux, built from two alphabets: the roots Ξ of one polynomial and the difference $\Xi - \mathbb{A}$ of the two root alphabets. We give three explicit forms for it: a Schur expansion whose coefficients are maximal minors of the band matrix of f ; a Jacobi–Trudi determinant with one alphabet per row, with the sign determined exactly; and an expansion in the basis $\{s_\nu(\Xi - \mathbb{A})\}$ whose coefficients are *skew* Schur functions of the roots of f , with an explicit vanishing criterion. For the principal coefficient of the classical subresultant the quotient collapses to a single supersymmetric Schur function indexed by the rectangle $(d_1 - k) \times (d_2 - k)$, recovering a theorem of Lascoux and Pragacz, and equivalently to a Toeplitz determinant in the Taylor coefficients of f/g , which is the Padé approximation form. We also show that the quotient is the determinant of divided differences of the rows, which settles the case of multiple roots at once. Finally we show that the naive multivariate analogue of the divisibility statement is false, and we identify the structural reason.

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1. INTRODUCTION

Let f, g be univariate polynomials of degrees d_1, d_2 over a field K , and let $\xi_1, \dots, \xi_{d_2}$ be the roots of g in $\overline{K}$. The classical Poisson product formula

$$\text{Res}(f, g) = (-1)^{d_1 d_2} b_{d_2}^{d_1} \prod_{j=1}^{d_2} f(\xi_j)$$

expresses the resultant as a polynomial expression in the roots of g and the coefficients of f . For subresultants the known analogues are not polynomial: Hong’s formula [5], and more generally the formulas of D’Andrea, Krick and Szanto [1], express subresultants as a *ratio* of two determinants, the denominator being the Vandermonde determinant $\mathcal{V}_{d_2} = \prod_{i < j} (\xi_j - \xi_i)$ of the roots of g . Concretely, [1, Thm. 2.2] states

$$\Delta_{\mathcal{S}} = \text{sg}(\mathcal{S}) b_{d_2}^{t^* - d_2 + 1} \frac{|\mathcal{O}_{\mathcal{S}}|}{\mathcal{V}_{d_2}}, \tag{1}$$

where $\mathcal{O}_{\mathcal{S}}$ is a $d_2 \times d_2$ matrix whose columns are indexed by the roots of g (all notation is recalled in Section 2).

This paper answers the following questions. Is the numerator $|\mathcal{O}_{\mathcal{S}}|$ always divisible by $\mathcal{V}_{d_2}$, so that (1) can be replaced by a formula which is universally polynomial in the roots? If so,

what is the quotient, and in which basis of symmetric functions is it naturally expressed? Do the answers extend to multiple roots and to the multivariate setting?

Results. Write

$$P_{\mathcal{S}} := \frac{|\mathcal{O}_{\mathcal{S}}|}{\mathcal{V}_{d_2}}.$$

- **Divisibility** (Theorem 3.2). Each column of $\mathcal{O}_{\mathcal{S}}$ depends on a single root, so $|\mathcal{O}_{\mathcal{S}}|$ vanishes on every diagonal $\xi_i = \xi_j$; since the roots of a generic g are algebraically independent, $\mathcal{V}_{d_2}$ divides $|\mathcal{O}_{\mathcal{S}}|$ in $\mathbb{Z}[a_0, \dots, a_{d_1}][\xi_1, \dots, \xi_{d_2}]$ and $P_{\mathcal{S}}$ is symmetric in the roots. The whole content of the problem is therefore the identification of $P_{\mathcal{S}}$.
- **Schur expansion** (Theorem 4.1). The matrix $\mathcal{O}_{\mathcal{S}}$ factors as $\rho \cdot W$, where W is the evaluation matrix of monomials at the roots and ρ is the coefficient matrix of the rows. The Cauchy–Binet formula and the bialternant formula give $P_{\mathcal{S}} = \sum_E \det(\rho_{\bullet, E}) s_{\lambda(E)}(\Xi)$: the quotient expands in the Schur basis of the roots of g with coefficients the maximal minors of the band matrix of f .
- **Multi-Schur form, with the exact sign** (Theorem 5.1). $P_{\mathcal{S}}$ is a Jacobi–Trudi determinant with *one alphabet per row*: the alphabet Ξ for the monomial rows and the difference alphabet $\Xi - \mathbb{A}$ for the rows coming from f . The global sign is $(-1)^{\binom{d_2}{2}} \text{sgn}(\sigma)$, and the statement holds for any ordering of the rows.
- **The classical case is a rectangle** (Theorem 6.1). When $\mathcal{S} = \{1, x, \dots, x^{k-1}\}$, that is for the principal coefficient of the classical subresultant, the determinant collapses and $P_{\mathcal{S}_k} = a_{d_1}^{d_2-k} s_{((d_1-k)d_2-k)}(\Xi - \mathbb{A})$: a single supersymmetric Schur function indexed by the rectangle $(d_1 - k) \times (d_2 - k)$. This recovers a theorem of Lascoux and Pragacz [6] from (1). The collapse uses in an essential way that the exponents of $\mathcal{S}$ are *consecutive*.
- **Toeplitz / Padé form** (Theorem 7.1). Equivalently, the principal coefficient is a Toeplitz determinant in the Taylor coefficients of $f^{\text{rev}}/g^{\text{rev}}$, which is the expansion of f/g at infinity. This is the subresultant–Padé dictionary.
- **The natural basis for arbitrary $\mathcal{S}$** (Theorem 8.1). For general $\mathcal{S}$ the quotient is *not* a single supersymmetric Schur function; the obstruction is caused by gaps in the exponent set. It does expand in that basis, by the Cauchy–Binet argument applied in the difference alphabet, and the resulting expansion is dual to the one of Theorem 4.1: the coefficients are *skew* Schur functions $s_{\lambda/\mu}(\mathbb{A})$ of the roots of f alone, and they are nonzero exactly when $\mu \subseteq \lambda$ and λ/μ has no column longer than d_1 .
- **Multiple roots** (Theorem 9.1). The quotient equals the determinant of the *divided differences* of the row polynomials, $P_{\mathcal{S}} = \det(R_i[\xi_1, \dots, \xi_j])$. This form is manifestly polynomial, and specializing repeated roots turns divided differences into derivatives; so the confluent case requires no new formula.
- **The multivariate case** (Proposition 10.1). The naive analogue of Theorem 3.2 is *false*. Already in the smallest example of [1, Ex. 3.4] the generalized Vandermonde determinant $\mathcal{V}_T$ divides none of the alternants occurring in the numerator. The reason is structural: the proof of Theorem 3.2 uses that alternating polynomials in one variable per point form a free module of rank one over the symmetric ones, and this fails for points of $\mathbb{A}^n$ with $n \geq 2$.

Status of the statements. Theorems 3.2, 4.1, 5.1, 6.1, 7.1, 8.1 and 9.1 are proved. All of them, together with Proposition 10.1, have in addition been verified by symbolic computation over a range of parameters; Section 11 reports the scope of those computations, which were also used to discover the statements.

2. SETUP AND NOTATION

We follow [1, §2]. Let

$$f = \sum_{l=0}^{d_1} a_l x^l, \quad g = \sum_{l=0}^{d_2} b_l x^l$$

be generic of degrees d_1, d_2 , with coefficients algebraically independent over $\mathbb{Q}$, and let $K = \mathbb{Q}(a_0, \dots, a_{d_1}, b_0, \dots, b_{d_2})$. Let $\Xi = \{\xi_1, \dots, \xi_{d_2}\}$ be the roots of g and $\mathbb{A} = \{\alpha_1, \dots, \alpha_{d_1}\}$ the roots of f in $\overline{K}$, and put

$$\mathcal{V}_{d_2} := \det(\xi_j^{i-1})_{1 \leq i, j \leq d_2} = \prod_{1 \leq i < j \leq d_2} (\xi_j - \xi_i).$$

Fix t with $0 \leq t \leq d_1 + d_2 - 1$ and set

$$t^* := \max\{d_2 - 1, t\}, \quad k := t + 1 - \max\{0, t - d_1 + 1\} - \max\{0, t - d_2 + 1\},$$

and let $\mathcal{S} = \{x^{\gamma_1}, \dots, x^{\gamma_k}\} \subset K[x]_t$ with $0 \leq \gamma_1 < \dots < \gamma_k \leq t$. Following [1], $\Delta_{\mathcal{S}}$ denotes the order t subresultant of f, g with respect to $\mathcal{S}$, and $\text{sg}(\mathcal{S})$ the sign defined there.

Definition 2.1. The matrix $\mathcal{O}_{\mathcal{S}} \in \overline{K}^{d_2 \times d_2}$ of [1, Thm. 2.2] has rows $(R_i(\xi_1), \dots, R_i(\xi_{d_2}))$, where the row polynomials R_i are

$$x^{\gamma_1}, \dots, x^{\gamma_k}, \quad x^{t+1}, \dots, x^{t^*}, \quad f, \quad xf, \quad \dots, \quad x^{t-d_1}f.$$

We call the first two groups the *monomial rows* and the last one the *f-rows*, and we write

$$\Gamma := \{\gamma_1, \dots, \gamma_k\} \cup \{t+1, \dots, t^*\}, \quad \Phi := \{d_1, d_1+1, \dots, t\}$$

for the sets of degrees of the two groups ($\Phi = \emptyset$ if $t < d_1$), and $\varepsilon_i := \deg R_i$, so that $\{\varepsilon_i\}_i = \Gamma \sqcup \Phi$ and $\#\Gamma + \#\Phi = d_2$. Finally let

$$\rho \in \mathbb{Z}[a_0, \dots, a_{d_1}]^{d_2 \times (t^*+1)}, \quad \rho_{i,c} := [x^c]R_i,$$

be the coefficient matrix of the rows: a unit row e_{γ} for each $\gamma \in \Gamma$, and the shifted rows $(a_0, \dots, a_{d_1})$ of the band matrix of f for the *f-rows*.

Throughout we set

$$P_{\mathcal{S}} := \frac{|\mathcal{O}_{\mathcal{S}}|}{\mathcal{V}_{d_2}}$$

so that (1) reads $\Delta_{\mathcal{S}} = \text{sg}(\mathcal{S}) b_{d_2}^{t^*-d_2+1} P_{\mathcal{S}}$.

The two alphabets. We use λ -ring notation for symmetric functions, as in [7, 6]. Let $h_r(\Xi)$ denote the complete homogeneous symmetric function, and let $h_r(\Xi - \mathbb{A})$ be defined by the generating series

$$\sum_{r \geq 0} h_r(\Xi - \mathbb{A}) z^r = \frac{\prod_{i=1}^{d_1} (1 - \alpha_i z)}{\prod_{j=1}^{d_2} (1 - \xi_j z)}.$$

Write $f^{\text{rev}}(z) := \sum_{r=0}^{d_1} a_{d_1-r} z^r$ and $g^{\text{rev}}(z) := \sum_{r=0}^{d_2} b_{d_2-r} z^r$, so that $\prod_j (1 - \xi_j z) = g^{\text{rev}}(z)/b_{d_2}$ and $\prod_i (1 - \alpha_i z) = f^{\text{rev}}(z)/a_{d_1}$. It is convenient to use the denominator-free normalizations

$$h_r := [z^r] \frac{1}{\prod_j (1 - \xi_j z)} = h_r(\Xi), \quad H_r := [z^r] \frac{f^{\text{rev}}(z)}{\prod_j (1 - \xi_j z)} = a_{d_1} h_r(\Xi - \mathbb{A}), \quad (2)$$

both of which lie in $\mathbb{Z}[a][\xi]$, with $h_r = H_r = 0$ for $r < 0$. We also put

$$c_r := [z^r] \frac{f^{\text{rev}}(z)}{g^{\text{rev}}(z)}, \quad \text{so that} \quad H_r = b_{d_2} c_r. \quad (3)$$

Note that $H_0 = a_{d_1}$, *not* 1; consequently a Jacobi–Trudi determinant of size L in the H_r carries a factor $a_{d_1}^L$, and the size of the determinant matters.

We grade $\mathbb{Z}[a][\xi]$ by

$$\deg \xi_j = 1, \quad \deg a_{d_1-r} = r. \quad (4)$$

With this grading h_r and H_r are homogeneous of degree r , the row R_i contributes degree ε_i , hence $|\mathcal{O}_S|$ is homogeneous of degree $\sum_i \varepsilon_i$ and $\mathcal{V}_{d_2}$ of degree $\binom{d_2}{2}$.

3. DIVISIBILITY

Lemma 3.1. *Let R be an integral domain and $F \in R[\xi_1, \dots, \xi_d]$ such that $F|_{\xi_i=\xi_j} = 0$ for all $i < j$. Then $\prod_{i < j} (\xi_j - \xi_i)$ divides F in $R[\xi_1, \dots, \xi_d]$.*

Proof. Since $\xi_j - \xi_i$ is monic of degree 1 in ξ_j , division with remainder by it is available over any commutative ring: $F = (\xi_j - \xi_i)Q + \varrho$ with ϱ not involving ξ_j beyond degree 0 in the division, and evaluating at $\xi_j = \xi_i$ gives $\varrho = 0$. Thus $\xi_j - \xi_i \mid F$ in $R[\xi]$. Write $F = (\xi_j - \xi_i)Q$. For any other pair $\{k, l\} \neq \{i, j\}$ we have $0 = F|_{\xi_k=\xi_l} = (\xi_j - \xi_i)|_{\xi_k=\xi_l} \cdot Q|_{\xi_k=\xi_l}$, and $(\xi_j - \xi_i)|_{\xi_k=\xi_l} \neq 0$ is a nonzerodivisor of the domain $R[\xi]$, so Q again vanishes on all the remaining diagonals. Iterating over the $\binom{d}{2}$ pairs gives the claim. $\square$

Theorem 3.2. $\mathcal{V}_{d_2}$ divides $|\mathcal{O}_S|$ in $\mathbb{Z}[a_0, \dots, a_{d_1}][\xi_1, \dots, \xi_{d_2}]$. The quotient P_S is symmetric in $\xi_1, \dots, \xi_{d_2}$, homogeneous of degree $t - d_1 + 1$ in the a_l , and homogeneous of degree $\sum_i \varepsilon_i - \binom{d_2}{2}$ for the grading (4).

Proof. Each column of $\mathcal{O}_S$ depends on a single root, so specializing $\xi_i = \xi_j$ produces two equal columns and $|\mathcal{O}_S|$ vanishes. The roots of a generic g are algebraically independent over $\mathbb{Q}$, so $|\mathcal{O}_S|$ is a polynomial in $\mathbb{Z}[a][\xi]$ vanishing on all diagonals, and Lemma 3.1 applies with $R = \mathbb{Z}[a_0, \dots, a_{d_1}]$. Transposing two roots transposes two columns of $\mathcal{O}_S$ and two factors of $\mathcal{V}_{d_2}$ with the same sign, so the quotient is symmetric. The degrees follow from the grading discussion after (4), each of the $t - d_1 + 1$ f -rows contributing degree 1 in the a_l . $\square$

Remark 3.3. By (1), $P_S = \pm b_{d_2}^{-(t^*-d_2+1)} \Delta_S$: the quotient is the subresultant, written in the roots of g and the coefficients of f , up to a power of the leading coefficient of g . The remaining sections identify it.

4. THE SCHUR EXPANSION

For $E = \{e_1 < \dots < e_{d_2}\} \subseteq \{0, 1, \dots, t^*\}$ define the partition

$$\lambda(E)_r := e_{d_2+1-r} - (d_2 - r), \quad r = 1, \dots, d_2. \quad (5)$$

Theorem 4.1. *With the notation of Definition 2.1,*

$$P_S = \sum_E \det(\rho_{\bullet, E}) s_{\lambda(E)}(\xi_1, \dots, \xi_{d_2}),$$

the sum being over all d_2 -subsets $E \subseteq \{0, \dots, t^\}$. Moreover $\det(\rho_{\bullet, E}) = 0$ unless $\Gamma \subseteq E$, in which case $\det(\rho_{\bullet, E})$ is, up to sign, the maximal minor of the band matrix of f on the columns $E \setminus \Gamma$.*

Proof. Let $W \in \overline{K}^{(t^*+1) \times d_2}$ be given by $W_{c,j} = \xi_j^c$. By Definition 2.1, $\mathcal{O}_S = \rho \cdot W$, and Cauchy–Binet gives $|\mathcal{O}_S| = \sum_E \det(\rho_{\bullet, E}) \det(W_{E, \bullet})$ over d_2 -subsets E taken in increasing order. The bialternant formula, in the increasing convention matching $\mathcal{V}_{d_2} = \det(\xi_j^{i-1})$, reads $\det(W_{E, \bullet}) = \mathcal{V}_{d_2} s_{\lambda(E)}(\Xi)$ with $\lambda(E)$ as in (5). Dividing by $\mathcal{V}_{d_2}$ gives the formula. The rows of ρ indexed by Γ are unit rows e_γ ; if $\gamma \notin E$ for some $\gamma \in \Gamma$, that row of $\rho_{\bullet, E}$ vanishes. Otherwise expanding along those rows leaves the minor of the f -band on the remaining columns. $\square$

Example 4.2. For $d_1 = d_2 = 4$, $t = 5$, $\mathcal{S} = \{1, x^2\}$:

$$\begin{aligned} P_{\mathcal{S}} = & (a_0 a_3 - a_1 a_2) s_{(0,0,0,0)} + (a_0 a_4 - a_1 a_3) s_{(1,0,0,0)} - a_1 a_4 s_{(2,0,0,0)} \\ & + (a_3^2 - a_2 a_4) s_{(1,1,1,0)} + a_3 a_4 s_{(2,1,1,0)} + a_4^2 s_{(2,2,1,0)}. \end{aligned}$$

The coefficients are 2×2 minors of the band matrix of f ; by the dual Jacobi–Trudi (Nägelsbach–Kostka) identity they are, up to a power of a_{d_1} and a sign, Schur functions of the roots of f . Theorem 8.1 makes this precise in the appropriate basis.

5. THE MULTI-SCHUR FORM AND THE EXACT SIGN

Theorem 5.1. *Let τ be any permutation of $\{1, \dots, d_2\}$ and let M^τ be the $d_2 \times d_2$ matrix*

$$M_{i,q}^\tau := X_{\varepsilon_{\tau(i)} - d_2 + q}(A_{\tau(i)}), \quad 1 \leq i, q \leq d_2,$$

where $X_r(A_i) = h_r$ if the row $\tau(i)$ is a monomial row and $X_r(A_i) = H_r$ if it is an f -row, with the convention $h_r = H_r = 0$ for $r < 0$. Then

$$P_{\mathcal{S}} = (-1)^{\binom{d_2}{2}} \operatorname{sgn}(\tau) \det M^\tau.$$

In particular, taking τ to order the rows by decreasing ε , the matrix M^τ is a Jacobi–Trudi matrix with one alphabet per row: the alphabet Ξ on the monomial rows and the difference alphabet $\Xi - \mathbb{A}$ on the f -rows.

Proof. The key observation is that $\mathcal{O}_{\mathcal{S}}$ and M^{id} have the same coefficient matrix ρ , applied to two different families of rows. Indeed, put $Y_{c,q} := h_{c-d_2+q}(\Xi)$ for $0 \leq c \leq t^*$ and $1 \leq q \leq d_2$. For a monomial row of degree γ we have $\rho_{i,\bullet} = e_\gamma$ and $(\rho \cdot Y)_{i,q} = h_{\gamma-d_2+q}$, as required. For the f -row $x^i f$ we have $\rho_{i,c} = a_{c-i}$, while by (2),

$$H_{(i+d_1)-d_2+q} = \sum_{l=0}^{d_1} a_l h_{i+l-d_2+q} = \sum_c \rho_{i,c} h_{c-d_2+q} = (\rho \cdot Y)_{i,q},$$

where we substituted $c = i + l$. Hence

$$\mathcal{O}_{\mathcal{S}} = \rho \cdot W, \quad M^{\text{id}} = \rho \cdot Y.$$

Applying Cauchy–Binet to both, with d_2 -subsets E taken in increasing order,

$$|\mathcal{O}_{\mathcal{S}}| = \sum_E \det(\rho_{\bullet,E}) \det(W_{E,\bullet}), \quad \det M^{\text{id}} = \sum_E \det(\rho_{\bullet,E}) \det(Y_{E,\bullet}).$$

As recalled in the proof of Theorem 4.1, $\det(W_{E,\bullet}) = \mathcal{V}_{d_2} s_{\lambda(E)}(\Xi)$. On the other hand $\det(Y_{E,\bullet})$ is a Jacobi–Trudi determinant for the same partition $\lambda(E)$, but with the rows in the opposite order: in the Jacobi–Trudi matrix of $\lambda(E)$ the row p carries the index e_{d_2+1-p} , that is the e 's in decreasing order, whereas $Y_{E,\bullet}$ lists them increasingly. Reversing d_2 rows contributes $(-1)^{\binom{d_2}{2}}$, so $\det(Y_{E,\bullet}) = (-1)^{\binom{d_2}{2}} s_{\lambda(E)}(\Xi)$. Comparing the two sums term by term yields $\det M^{\text{id}} = (-1)^{\binom{d_2}{2}} |\mathcal{O}_{\mathcal{S}}| / \mathcal{V}_{d_2}$. Finally $\det M^\tau = \operatorname{sgn}(\tau) \det M^{\text{id}}$, and the sign is its own inverse. $\square$

Remark 5.2. The proof does not use the ordering of the rows, which is why the statement holds for an arbitrary τ : interchanging two rows changes $\operatorname{sgn}(\tau)$ and $\det M^\tau$ simultaneously. In particular ties among the ε_i — which occur whenever a monomial exponent equals $i + d_1$ — are harmless.

6. THE CLASSICAL CASE: A RECTANGLE

We now specialize to the classical scalar subresultants. By [1, Rmk. 2.1(2)], for $0 \leq k \leq \min\{d_1, d_2\}$ and $t = d_1 + d_2 - k - 1$ one takes $\mathcal{S}_j := \{x^i : 0 \leq i \leq k, i \neq j\}$, and then $\Delta_{\mathcal{S}_j} = (-1)^{(d_1-k)(d_2-k)} S_k^{(j)}$, where $S_k^{(j)}$ is the scalar subresultant of [1, (3)]. Note that $t \geq d_2 - 1$, so $t^* = t$ and there are no rows $x^{t+1}, \dots, x^{t^*}$; moreover $\Gamma = \{0, \dots, k\} \setminus \{j\}$ and $\Phi = \{d_1, \dots, t\}$ has $m := d_2 - k$ elements.

Theorem 6.1. *Let $\mathcal{S}_k = \{1, x, \dots, x^{k-1}\}$, i.e. $j = k$, and $m = d_2 - k$. Then*

$$P_{\mathcal{S}_k} = \det \left(H_{(d_1-k)-p+q} \right)_{p,q=1}^m = a_{d_1}^{d_2-k} s_{((d_1-k)d_2-k)}(\Xi - \mathbb{A}),$$

with sign $+1$: the supersymmetric Schur function of the rectangular partition $(d_1 - k) \times (d_2 - k)$ in the difference alphabet.

Proof. Here $\Gamma = \{0, 1, \dots, k-1\}$ and the degrees are $\varepsilon = (0, 1, \dots, k-1, d_1, d_1+1, \dots, d_1+m-1)$, already increasing. Ordering by decreasing ε is therefore the full reversal of d_2 rows, whose sign $(-1)^{\binom{d_2}{2}}$ cancels the one in Theorem 5.1; so $P_{\mathcal{S}_k} = \det M^\tau$ with M^τ the Jacobi–Trudi matrix in decreasing order. Write M^τ in blocks, with the m f -rows first (row $p = 1, \dots, m$ carrying $\varepsilon = d_1 + m - p$) and the k monomial rows below (row $p' = 1, \dots, k$ carrying $\gamma = k - p'$), and split the columns as $q \leq m$ and $q = m + r$ with $1 \leq r \leq k$.

Lower left block ($k \times m$) vanishes. Its entry is $h_{\gamma-d_2+q}$ with $\gamma \leq k-1$ and $q \leq m = d_2 - k$, so $\gamma - d_2 + q \leq -1 < 0$.

Lower right block ($k \times k$) is unitriangular. With $q = m + r$ its entry is $h_{(k-p')-d_2+(d_2-k+r)} = h_{r-p'}$, which is upper triangular with $h_0 = 1$ on the diagonal; its determinant is 1.

Upper left block ($m \times m$). Its entry is $H_{(d_1+m-p)-d_2+q} = H_{(d_1-k)-p+q}$, because $m = d_2 - k$.

Being block upper triangular, $\det M^\tau$ is the product of the two diagonal blocks, i.e. the Jacobi–Trudi determinant of the rectangle $((d_1 - k)^m)$ in the H 's. By (2) each of the m rows carries a factor a_{d_1} , giving the stated normalization. $\square$

Corollary 6.2. $S_k^{(k)} = (-1)^{(d_1-k)(d_2-k)} b_{d_2}^{d_1-k} a_{d_1}^{d_2-k} s_{((d_1-k)d_2-k)}(\Xi - \mathbb{A})$.

Proof. Combine Theorem 6.1 with (1) and $\Delta_{\mathcal{S}_k} = (-1)^{(d_1-k)(d_2-k)} S_k^{(k)}$, using $t^* - d_2 + 1 = d_1 - k$ and $\text{sg}(\mathcal{S}_k) = +1$; the latter because for $\mathcal{S}_k = \{1, \dots, x^{k-1}\}$ with $t^* = t$ the reordering defining sg in [1, §2] is the identity. $\square$

Corollary 6.2 is the statement of Lascoux and Pragacz [6] for the principal subresultant coefficient, here obtained from (1). The extreme cases are consistent: $k = 0$ gives the rectangle $d_1 \times d_2$, i.e. the resultant, and $k = \min\{d_1, d_2\}$ gives the empty partition.

Remark 6.3. The proof uses in an essential way that the monomial exponents $\Gamma = \{0, \dots, k-1\}$ are *consecutive*: this is what makes the lower left block vanish and the lower right block unitriangular. For $j < k$ the set $\{0, \dots, k\} \setminus \{j\}$ has a gap and the collapse fails. For instance for $d_1 = d_2 = 3$, $k = 2$, $j = 0$ one gets

$$P_{\mathcal{S}_0} = a_0 + a_3 e_3(\Xi) = a_3(e_3(\Xi) - e_3(\mathbb{A})),$$

which is the mixed determinant

$$\det \begin{pmatrix} H_1 & H_2 & H_3 \\ 1 & h_1 & h_2 \\ 0 & 1 & h_1 \end{pmatrix},$$

that is, the truncated alternating sum $\sum_{i=0}^2 (-1)^i H_{3-i} e_i(\Xi)$; the untruncated sum would collapse to $[z^3] f^{\text{rev}}(z) = a_0$. An exhaustive search over partitions confirms that $P_{\mathcal{S}_j}$ is not a single supersymmetric Schur function when $j < k$.

7. TOEPLITZ DETERMINANTS AND PADÉ APPROXIMATION

Theorem 7.1. *With c_r as in (3) and $m = d_2 - k$,*

$$S_k^{(k)} = (-1)^{(d_1-k)(d_2-k)} b_{d_2}^{d_1+d_2-2k} \det (c_{(d_1-k)-p+q})_{p,q=1}^m.$$

Proof. By (3), $H_r = b_{d_2} c_r$, so the $m \times m$ determinant of Theorem 6.1 equals $b_{d_2}^m \det(c_{(d_1-k)-p+q})$. Now apply Corollary 6.2, or equivalently multiply by $b_{d_2}^{d_1-k}$ as in its proof; $m + (d_1 - k) = d_1 + d_2 - 2k$. $\square$

Since c_r are the Taylor coefficients of $f^{\text{rev}}/g^{\text{rev}}$, i.e. of the expansion of f/g at infinity, Theorem 7.1 says that the principal subresultant coefficient is a Toeplitz determinant of that series. This is exactly the determinantal criterion underlying the Padé approximation problem, and it makes explicit the link with the rational interpolation viewpoint of [3].

8. THE NATURAL BASIS FOR ARBITRARY $\mathcal{S}$

By Remark 6.3 the quotient is not a single supersymmetric Schur function in general. It does, however, expand in the basis $\{s_\nu(\Xi - \mathbb{A})\}$, and the expansion is dual to that of Theorem 4.1. Throughout this section we normalize f to be monic, $a_{d_1} = 1$, so that the $h_s(\mathbb{A})$ are polynomials in the a_l ; in general they lie in $\mathbb{Z}[a][a_{d_1}^{-1}]$.

Set $u_r := h_r(\Xi - \mathbb{A})$ and $U_{r,q} := u_{r-d_2+q}$.

Theorem 8.1. *Let $\mathcal{C}$ be the matrix whose rows are indexed as in Definition 2.1, with*

$$\mathcal{C}_{i,c} = \delta_{c,\varepsilon_i} \text{ for an } f\text{-row}, \quad \mathcal{C}_{i,c} = h_{\gamma-c}(\mathbb{A}) \text{ for a monomial row of degree } \gamma.$$

Then $M^{\text{id}} = \mathcal{C} \cdot U$ and consequently

$$P_{\mathcal{S}} = (-1)^{\binom{d_2}{2}} \sum_E \det(\mathcal{C}_{\bullet,E}) \det(U_{E,\bullet}) = \sum_F \pm \det(\mathcal{C}_{\bullet,\Phi \sqcup F}) s_{\nu(\Phi \sqcup F)}(\Xi - \mathbb{A}),$$

where F runs over the $\#\Gamma$ -subsets of $\mathbb{Z}_{\geq 0} \setminus \Phi$ and $\nu(\cdot)$ is as in (5). Moreover, taking the rows (indexed by Γ) and the columns (indexed by F) in decreasing order,

$$\det(\mathcal{C}_{\bullet,\Phi \sqcup F}) = \pm \det(h_{\gamma_p - f_q}(\mathbb{A}))_{p,q} = \pm s_{\lambda(\Gamma)/\mu(F)}(\mathbb{A}), \quad \lambda(\Gamma)_p = \gamma_p + p, \quad \mu(F)_q = f_q + q.$$

Proof. Since $\Xi = (\Xi - \mathbb{A}) + \mathbb{A}$ we have $h_r(\Xi) = \sum_s h_s(\mathbb{A}) h_{r-s}(\Xi - \mathbb{A})$, which is exactly the monomial row of $\mathcal{C}$ applied to U ; and $H_r = a_{d_1} u_r = u_r$ for monic f , which is the unit row. Hence $M^{\text{id}} = \mathcal{C} \cdot U$, and Cauchy–Binet together with Theorem 5.1 gives the first display, the two occurrences of $(-1)^{\binom{d_2}{2}}$ cancelling exactly as in the proof of Theorem 5.1. The unit rows force $\Phi \subseteq E$, whence the reindexing $E = \Phi \sqcup F$. Finally, the surviving minor is $\det(h_{\gamma_p - f_q}(\mathbb{A}))$, and with both indices decreasing the sequences $\lambda_p = \gamma_p + p$ and $\mu_q = f_q + q$ are weakly decreasing, so this is the skew Jacobi–Trudi determinant of $\lambda(\Gamma)/\mu(F)$, cf. [8, I.(5.4)]. $\square$

Corollary 8.2. *$\det(\mathcal{C}_{\bullet,\Phi \sqcup F}) \neq 0$ if and only if $\mu(F) \subseteq \lambda(\Gamma)$ and the skew diagram $\lambda(\Gamma)/\mu(F)$ has no column of length greater than d_1 .*

Proof. $s_{\lambda/\mu} = 0$ unless $\mu \subseteq \lambda$, and a skew Schur function vanishes on an alphabet of d_1 letters exactly when the skew diagram has a column of length $> d_1$; see [8, I.5]. Here $\#\mathbb{A} = d_1$. $\square$

Corollary 8.2 explains the sparsity of the expansion and makes precise the role of the gaps of Γ : for $\Gamma = \{k-1, \dots, 0\}$ one gets $\lambda(\Gamma)_p = (k-p) + p = k$, i.e. λ is the $k \times k$ square, and a single F survives, recovering Theorem 6.1.

Example 8.3. With f monic, the expansions are short. All Schur functions below are taken in the alphabet $\Xi - \mathbb{A}$:

$$\begin{aligned} d_1 = 3, d_2 = 3, t = 4, \mathcal{S} = \{x\} : P_{\mathcal{S}} &= -a_2 s_{(2,2)} + s_{(2,2,1)}, \\ d_1 = 2, d_2 = 3, t = 3, \mathcal{S} = \{x^2\} : P_{\mathcal{S}} &= (a_1^2 - a_0) s_{(1,1)} - a_1 s_{(1,1,1)}, \\ d_1 = 4, d_2 = 4, t = 5, \mathcal{S} = \{1, x^2\} : P_{\mathcal{S}} &= -a_3 s_{(2,2)} + s_{(2,2,1)}, \\ d_1 = 3, d_2 = 3, t = 3, \mathcal{S} = \{1, x\} : P_{\mathcal{S}} &= -s_{(1)}. \end{aligned}$$

The last line has a single term, as it must: there $\Gamma = \{0, 1\}$ is an interval. The coefficients are the $h_s(\mathbb{A})$: for monic f , $h_1(\mathbb{A}) = -a_{d_1-1}$, $h_2(\mathbb{A}) = a_{d_1-1}^2 - a_{d_1-2}$, and so on.

9. MULTIPLE ROOTS: THE NEWTON FORM

If g has repeated roots both $|\mathcal{O}_{\mathcal{S}}|$ and $\mathcal{V}_{d_2}$ vanish and (1) is a $0/0$ expression. The following identity removes the difficulty at once. Recall that the divided difference of a polynomial R at nodes $\xi_1, \dots, \xi_j$ is $R[\xi_1, \dots, \xi_j] = \sum_{i=1}^j R(\xi_i) / \prod_{l \neq i} (\xi_i - \xi_l)$.

Theorem 9.1. *With the row polynomials R_i of Definition 2.1,*

$$P_{\mathcal{S}} = \det (R_i[\xi_1, \dots, \xi_j])_{1 \leq i, j \leq d_2}.$$

Proof. Let T be the $d_2 \times d_2$ matrix expressing the divided differences in terms of the evaluations, $T_{j,j'} = 1 / \prod_{l \leq j, l \neq j'} (\xi_{j'} - \xi_l)$ for $j' \leq j$ and $T_{j,j'} = 0$ for $j' > j$. It is lower triangular with diagonal entries $T_{j,j} = 1 / \prod_{l < j} (\xi_j - \xi_l)$, so $\det T = \prod_{l < j} (\xi_j - \xi_l)^{-1} = \mathcal{V}_{d_2}^{-1}$. Since $(R_i[\xi_1, \dots, \xi_j])_{i,j} = \mathcal{O}_{\mathcal{S}} \cdot T^t$, taking determinants gives the claim. $\square$

Corollary 9.2. *Let g have distinct roots $y_1, \dots, y_s$ with multiplicities $m_1, \dots, m_s$, $\sum_i m_i = d_2$. Let $\mathcal{O}_{\mathcal{S}}^{\text{conf}}$ be the confluent matrix whose columns are indexed by the pairs (i, r) , $0 \leq r < m_i$, with entries $\frac{1}{r!} R^{(r)}(y_i)$, and let $\mathcal{V}^{\text{conf}} = \prod_{i < i'} (y_{i'} - y_i)^{m_i m_{i'}}$. Then*

$$\frac{|\mathcal{O}_{\mathcal{S}}^{\text{conf}}|}{\mathcal{V}^{\text{conf}}} = P_{\mathcal{S}}(y_1, \dots, y_1, \dots, y_s, \dots, y_s),$$

each y_i repeated m_i times. In particular the case of multiple roots needs no new formula: the universal polynomial $P_{\mathcal{S}}$ evaluated at the multiset of roots computes it.

Proof. Split the roots into s groups, the i -th consisting of $\xi_{i,1}, \dots, \xi_{i,m_i}$, and let $\mathcal{V}^{(i)} := \prod_{l < j} (\xi_{i,j} - \xi_{i,l})$ be the Vandermonde determinant of the i -th group. Performing inside each group the column operations of Theorem 9.1, which replace the columns $R(\xi_{i,1}), \dots, R(\xi_{i,m_i})$ by the divided differences

$$R[\xi_{i,1}], \dots, R[\xi_{i,1}, \dots, \xi_{i,m_i}],$$

multiplies the determinant by $\prod_i (\mathcal{V}^{(i)})^{-1}$. Hence

$$\frac{|\mathcal{O}_{\mathcal{S}}|}{\prod_i \mathcal{V}^{(i)}} = \det \left(\text{within-group divided differences} \right),$$

and on the other hand, splitting the factors of $\mathcal{V}_{d_2}$ according to whether the two roots lie in the same group or not,

$$\frac{\mathcal{V}_{d_2}}{\prod_i \mathcal{V}^{(i)}} = \prod_{i < i'} \prod_{j, j'} (\xi_{i',j'} - \xi_{i,j}).$$

Now let $\xi_{i,j} \rightarrow y_i$ for every j . The left-hand sides converge to $|\mathcal{O}_S^{\text{conf}}|$ and $\mathcal{V}^{\text{conf}}$ respectively, because divided differences at coalescing nodes converge to the divided derivatives, $R[\underbrace{y, \dots, y}_{r+1}] =$

$R^{(r)}(y)/r!$, and because the second product converges to $\prod_{i < i'} (y_{i'} - y_i)^{m_i m_{i'}}$. Since $P_S = |\mathcal{O}_S|/\mathcal{V}_{d_2}$ is the ratio of the two left-hand sides and is a polynomial, its value at the limit is the limit of the values, which gives the statement. $\square$

10. THE MULTIVARIATE CASE: AN OBSTRUCTION

In the multivariate setting [1, Thm. 3.2] the denominator is the generalized Vandermonde determinant $\mathcal{V}_T = \det(\xi_j^{\alpha_i})$ built on the $\mathbf{d} = d_1 \cdots d_n$ common roots $\xi_j \in \overline{K}^n$ of $f_1, \dots, f_n$. The analogue of Theorem 3.2 fails.

Proposition 10.1. *Let $n = 2$, $d_1 = d_2 = d_3 = 2$ and $t = 2$ as in [1, Ex. 3.4], so that*

$$|\mathcal{O}_S| = -c_0|V_{2,5}| + c_2|V_{0,5}| + c_5|V_{0,2}|, \quad \mathcal{V}_T = |V_{4,5}|,$$

where V is the 6×4 matrix of evaluations of $1, u, v, uv, u^2, v^2$ at the four roots $\xi_j = (u_j, v_j)$ and $V_{i,i'}$ denotes the 4×4 submatrix obtained by deleting rows i, i' . Then $\mathcal{V}_T$ divides none of $|V_{2,5}|, |V_{0,5}|, |V_{0,2}|$ in $\mathbb{Q}[u_1, \dots, u_4, v_1, \dots, v_4]$, and hence $\mathcal{V}_T \nmid |\mathcal{O}_S|$.

Proof (computational). The coefficients c_0, c_2, c_5 of f_3 are independent indeterminates, so $\mathcal{V}_T$ divides $|\mathcal{O}_S|$ if and only if it divides each of the three alternants separately. A direct division in $\mathbb{Q}[u, v]$ shows that it divides none of them; in fact none of the 14 determinants $|V_{i,i'}|$ with $(i, i') \neq (4, 5)$ is divisible by $\mathcal{V}_T$. The test is conclusive because here the roots are free parameters: four general points of the plane are the base locus of a pencil of conics, so the map from coefficients to roots is dominant and the eight coordinates u_j, v_j are algebraically independent; therefore “divisible in $\mathbb{Q}[u, v]$ ” is the same as “universally polynomial in the roots”. The degrees are $\deg \mathcal{V}_T = 4$ and $\deg |V_{2,5}| = 5$, $\deg |V_{0,5}| = 6$, $\deg |V_{0,2}| = 7$. $\square$

The reason is structural. The proof of Theorem 3.2 rests on Lemma 3.1, i.e. on the fact that in $K[\xi_1, \dots, \xi_d]$ with scalar ξ_j the module of alternating polynomials is free of rank one over the symmetric ones, generated by the Vandermonde determinant. For points of $\mathbb{A}^n$ with $n \geq 2$ the corresponding module of diagonally alternating polynomials is not principal — this is the setting of diagonal harmonics, cf. [11] — and $\mathcal{V}_T$ is merely one of several generators. Consequently the univariate statement cannot be transcribed, and the multivariate problem must be reformulated. Natural reformulations are: to exhibit explicitly the cancellation of $\mathcal{V}_T$ against the product $\prod_j \overline{\Delta}_{T_j}$ of homogeneous subresultants appearing in [1, (13)], which is polynomial in the coefficients; to determine the minimal universal denominator, i.e. $\gcd(|\mathcal{O}_S|, \mathcal{V}_T)$; or to expand $|\mathcal{O}_S|$ in a generating set of the module of alternating polynomials. We leave this for future work.

11. SYMBOLIC VERIFICATION

All statements above were verified by symbolic computation, with two independent implementations (SymPy 1.14 and SageMath 10.7); the Schur expansions were cross-checked against the native symmetric function routines of SageMath. We record the scope.

Statement	Range verified
Thms. 3.2, 4.1, 5.1	12 configurations $(d_1, d_2, t, \mathcal{S})$ with $d_1, d_2 \leq 6$, including $t < d_2 - 1$ and ties among the ε_i ; among them [1, Ex. 2.3 and 2.4].
Thm. 6.1	all $1 \leq d_1, d_2 \leq 5$ and all $0 \leq k \leq \min\{d_1, d_2\}$, without search.
Thm. 7.1	46 cases, $1 \leq d_1, d_2 \leq 4$ and all k , with a and b independent (not passing through the roots).
Thm. 8.1, Cor. 8.2	5 configurations; the vanishing criterion was tested on <i>all</i> 36 admissible subsets F .
Thm. 9.1	6 configurations; Cor. 9.2 on 8 multiplicity patterns $[2], [2, 1], [3], [2, 2], [3, 1], [2, 1, 1]$.
Prop. 10.1	all 15 alternants of [1, Ex. 3.4].

Finally, everything above is relative to (1). As an end-to-end check we also verified, for 20 tuples (d_1, d_2, k, j) , the identity

$$S_k^{(j)} = (-1)^{(d_1-k)(d_2-k)} \operatorname{sg}(\mathcal{S}_j) b_{d_2}^{d_1-k} P_{\mathcal{S}_j}$$

with $S_k^{(j)}$ computed from its classical determinantal definition [1, (3)] and b_l rewritten in the roots as $b_l = b_{d_2}(-1)^{d_2-l} e_{d_2-l}(\Xi)$, *with the exact signs*, including $\operatorname{sg}(\mathcal{S}_j)$. This is the only check that does not take (1) as a hypothesis, and it validates it as well.

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